

Deepa Shree Chickballapur Venkatachalapathi

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EDUCATION

University of Wisconsin - Madison

Master of Science in Data Science, GPA: 3.67/4.0

Madison, WI

Sep 2024 - May 2026

- **Awards:** Merit-Based Scholarship, Founder Lab – Selected for pilot-ready product cohort
- **Coursework:** Master's Research in Databases, Database Design & Implementation, Statistical Methods, Optimization, Machine Learning, Advanced Deep Learning

PES University

Bachelor of Technology in Computer Science and Engineering, GPA: 9.24/10.0

Bengaluru, India

Aug 2019 - May 2023

- **Awards:** Prof. MRD Scholarship (Top 5% students)
- **Coursework:** Data Structures, Linear Algebra, Operating Systems, Cloud Computing, Big Data, Distributed Systems

EXPERIENCE

Bank OZK

Data Engineer

Little Rock, AR

Jun 2026 – Present

- Incoming Data Engineer in the Risk Management Team.

Amazon Web Services

Software Development Engineer Intern

Seattle, WA

May 2025 – Aug 2025

- Developed ETL pipelines and data models for **terabyte-scale** AWS Host vulnerability data using SQL, PySpark & AWS services (S3, Glue, Redshift, Athena).
- Automated infrastructure & workflows with CloudFormation, AWS secure-cdk & Typescript for scalable data processing.
- Built near real-time dashboards with CloudWatch, SNS, and Lambda, improving host **security visibility** by **10%**.

University of Wisconsin-Madison

Graduate Research Assistant

Madison, USA

Feb 2025 – May 2025

- Fine-tuned vision models in PyTorch on HTC clusters to classify cranberry growth stages for accurate nutrient supply.
- Used roboflow to automate annotation of cranberries to support object detection to improve prediction **accuracy** by **7%**.

Hewlett Packard Enterprise

Cloud Developer - I

Bengaluru, India

Aug 2023 – Aug 2024

- Optimized metadata retrieval with natural language support for CMF (MLOps library) with RAG Pipelines using Python, Hugging Face and Pinecone.
- Developed a voice-based UI navigation feature, integrating STT & TTS with BERT using React.js & Python.
- Increased **impression rate** by **23%** with UI and dashboard development using React.js, Grafana and Humio.
- Reduced database **fetch time** by **80%** with caching optimization using Python, PostgreSQL, and Redis.
- Automated error logging pipelines using GitHub Actions and Terraform, reducing **latency** by **90%**.

Research and Development Intern

Jan 2023 – July 2023

- Automated regression testing using JUnit & Jenkins, reducing manual effort and improving **deployment speed** by **30%**.
- Built an NLP-based streaming log parser in Python to analyze gigabyte-scale SFTP logs for rapid issue detection.
- Reduced **triage time** with the parser by **80%**, enhancing reliability and maintainability of production systems.

Adobe Systems

Technical Consultant Intern

Bengaluru, India

May 2022 – Aug 2022

- Incremental ML pipeline for customer purchase propensity prediction with **F-measure over 0.9** using Python.
- Implemented interactive dashboards using MySQL and Power BI, increasing **reporting efficiency** by **15%**.

PROJECTS

Intelligent Pricing and Discount System | Python, Kafka, Docker, SQL, sklearn, FastAPI | Sponsor: Greenheck Group

- Built an AI-driven pricing and discount recommendation system to assist Greenheck's product managers.

Automating Analysis for Data Cubes | Databases, SQL, C++ | Advisor: Prof Goetz Graefe

- Designed optimized SQL and indexing for cost efficient incremental maintenance of data cubes and materialized views.

TECHNICAL SKILLS

Programming Languages & Scripting: Python, C++ (Proficient); JavaScript, Java, C, R, Bash, Linux CLI (Intermediate)
Data Management & Analysis: SQL, MongoDB, Tableau, PowerBI, PySpark, Hadoop, Kafka, dbt, Airflow, ElasticSearch
Machine Learning : Tensorflow, Keras, PyTorch, HuggingFace, LangChain, Pinecone, NLTK, scikit-learn, pandas, numpy
Cloud, Development & Monitoring Tools: Docker, Kubernetes, AWS, React.js, Git, Github Actions, Terraform, Grafana